ABSTRACT

The Drug Inventory and Supply Chain Tracking System using Hyperledger Fabric is designed to improve transparency, traceability, and security in pharmaceutical distribution. The proposed system uses blockchain technology to maintain a tamper-resistant record of drug registration, ownership transfers, and verification throughout the supply chain. Each medicine is registered with a unique serial number, batch number, drug name, manufacturer, manufacturing date, expiry date, and quantity. Hyperledger Fabric smart contracts validate transactions and maintain consistent drug records across authorized participants in the network. A web-based interface enables users to register medicines, view inventory details, transfer ownership, verify drug authenticity, and inspect transaction history. QR codes containing unique drug serial numbers support quick identification and verification through a connected application. A Node.js and Express backend communicates with the Hyperledger Fabric network, while the frontend provides an accessible interface for managing pharmaceutical records. By recording supply chain activities on a permissioned blockchain, the system helps identify unauthorized modifications, improves accountability, and supports the detection of counterfeit or unverified medicines. The proposed framework provides a secure and transparent approach to pharmaceutical inventory management and supply chain tracking.

Keywords: Hyperledger Fabric, Blockchain, Drug Traceability, Pharmaceutical Supply Chain, Smart Contracts, Drug Verification, QR Code, Inventory Management, Transaction History, Counterfeit Drug Detection.

The system also maintains a chronological history of drug-related transactions, allowing authorized users to examine ownership changes and track medicine movement through the supply chain. Smart contracts enforce the defined rules for registering drugs, retrieving records, verifying authenticity, and transferring ownership. The integration of a web application and an Android-based verification interface improves accessibility and simplifies drug identification. By combining permissioned blockchain infrastructure, unique drug identifiers, and QR-based verification, the proposed system aims to strengthen supply chain transparency, reduce the risk of unauthorized record manipulation, and support more reliable pharmaceutical distribution